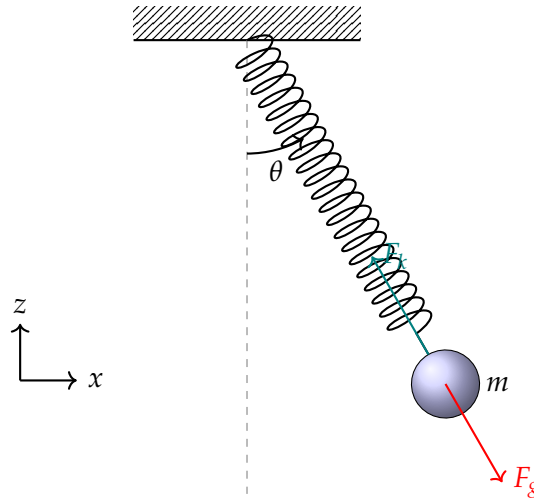


Spring Pendulum: The Theory of Hybrid Systems

MECH0107 Coursework 1

Data-Driven Methods for Engineers

March 13, 2026



Submission Note:

This report contains the analysis of the hybrid mass-spring-pendulum system using Computer Vision, Time-Frequency Analysis, and Dimensionality Reduction (PCA/SVD).

AI Usage Declaration (Category 2 – Assistive Role):

In accordance with the module's AI usage policy, generative AI tools were used solely in an assistive capacity during the preparation of this coursework. Specifically, Claude (Anthropic) was used to check grammar and spelling in the written report, to help interpret error messages encountered during code development, and to assist in creating TikZ diagrams for the L^AT_EX report. No AI tools were used for code writing or to generate any analytical ideas related to the coursework.

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1 Introduction and Problem Statement

Dynamical systems, which characterise the evolution of a state over time, are widely used to model real-world phenomena. When analysing these systems, one key challenge is to extract and interpret a system's behaviour by analysing the trajectory and solutions to the governing equations of their motion[1]. The elastic pendulum is a canonical example of a hybrid dynamical system, one that combines multiple modes of motion into a single framework[2]. The dynamics of the spring-mass system follow Hooke's Law ($F = -kx$), providing a restoring force F acting in the direction of equilibrium and proportional to the displacement x . Conversely, the pendulum's motion is governed by gravity ($g = 9.81 \text{ m/s}^2$), pulling the mass towards its vertical equilibrium. Under the small-angle approximation, which assumes small angular displacement (θ) and system decoupling, the two linearised equations are:

$$\underbrace{\ddot{\theta} + \frac{g}{L}\theta}_{\text{pendulum}} = 0 \quad (1)$$

$$\underbrace{\ddot{z} + \frac{k}{m}z}_{\text{spring}} = 0 \quad (2)$$

where L is the length of the spring, k is the spring constant, and m is the object's mass. **Equation (1)** describes the pendulum's oscillation in the horizontal, $x -$ plane, and **Equation (2)** refers to the spring's mode along the vertical $y -$ plane.

The coupling of the elastic and gravitational dynamics introduces an additional degree of freedom relative to either subsystem, producing a complex, non-linear system that exhibits phase coupling and energy exchange between modes[3]. This non-linearity, arises at large amplitudes which cause the small-angle approximation to break down, making the elastic pendulum a worthy benchmark for data-driven analysis of real-world non-linear systems.

This project analyses a hybrid mass-spring-pendulum system, released from an off-centre position, inducing both vertical and horizontal oscillations. The motion, captured across three cameras positioned at varied points around the system, is stored in an individual MATLAB (.mat) file, containing the raw image frames from that camera's viewpoint. The primary objective is to identify the dominant modes of motion, estimate the underlying physics and assess the validity of a simple analytical model against the experimental data.

The planned workflow was fourfold. First, the raw data is pre-processed, extracting the object's trajectory. Second, time-frequency analysis is applied to determine how the dominant frequency components evolve over time. Third, dimensionality reduction via PCA is applied to the combined, multi-camera dataset to identify the dominant modes of motion. Finally, an analytical model derived from the small-angle approximation is compared against the experimentally identified frequencies and modes of motion (see **Figure 1** for a detailed breakdown of the individual steps).

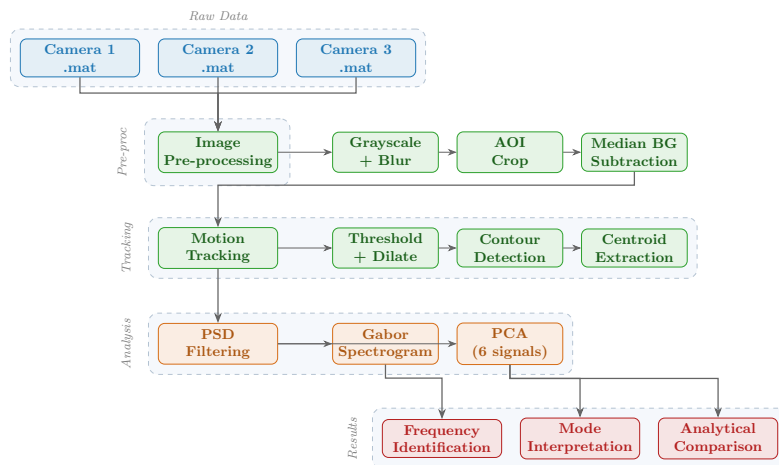


Figure 1: Planned project workflow and data pipeline.

2 Methodology

2.1 Image Pre-Processing

To extract the trajectory of the object, several pre-processing steps were utilised. First, each camera frame was converted from 4D arrays of shape $[i, j, c, t]$, corresponding to height, width, RGB channels and frame index to greyscale by averaging the three colour channels—yielding a 3-D array of shape $[i, j, t]$. Prior to this step, Camera 3 was rotated to align its vertical axis with Camera 1 and 2.

$$G[i, j, t] = \frac{1}{3} \sum_{c=1}^3 F[i, j, c, t] \quad (3)$$

To reduce the noise in each frame, a Spatial Gaussian Blur (also known as a low-pass filter) was applied to each frame. This method allowed for principal control over the filter's scaling and can be thought of as performing a convolution of an image using a 2-D Gaussian kernel, $h[u, v]$:

$$h[u, v] = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{u^2 + v^2}{2\sigma^2}\right), \quad u, v \in \{-r, \dots, r\}, \quad r = \frac{K-1}{2}, \quad (4)$$

where u is the distance from the origin (centre of the image) on the horizontal axis, v is the distance from the origin on the vertical axis, r is the kernel half-width, and σ is the standard deviation of the Gaussian distribution. For this experiment, the kernel size, K was set to 15x15 (15), yielding a σ of 2.6 (computed via OpenCV's default relation). The Gaussian kernel $h[u, v]$ represents a grid of weights that are highest at the centre and taper off according to σ . This process was applied to $G[i, j, t]$ with the filtered output shown in Equation (5).

$$\tilde{G}[i, j, t] = \sum_{u=-r}^r \sum_{v=-r}^r h[u, v] G[i - u, j - v, t], \quad (5)$$

Following low-pass filtering, each frame was spatially filtered (cropped) to the camera's associated Area of Interest (AOI). By localising the oscillating mass, this step improved algorithmic robustness by reducing the total number of pixels processed in subsequent steps. These dimensions were adjusted manually to create an optimal boundary region:

$$\tilde{G}_c^{\text{AOI}}[i, j, t] = \tilde{G}_c[i + i_{c,\min}, j + j_{c,\min}, t] \quad (6)$$

for (i, j) inside the AOI bounds. Where $c \in \{1, 2, 3\}$ represents camera index and $t \in \{0, 1, \dots, n-1\}$ the frame index (n is the maximum number of frames for the respective camera).

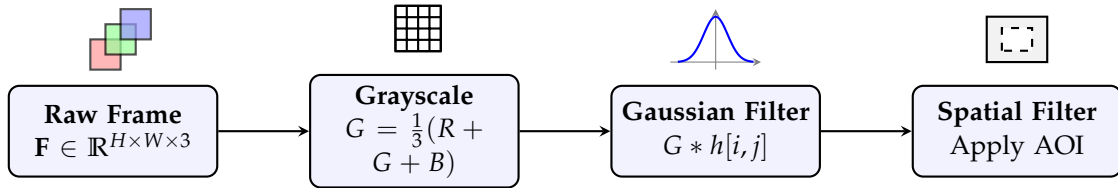


Figure 2: Image pre-processing pipeline

2.2 Motion Tracking

To determine the precise coordinates of the oscillating mass, a background subtraction pipeline was implemented (Figure 3). Background subtraction was first utilised to isolate the oscillating object from the static 8-bit background. By subtracting the current frame ($\tilde{G}_c^{\text{AOI}}[i, j, t]$) from the background ($B_c[i, j]$), a foreground mask was generated ($D_{c,t}[i, j]$) [4].

$$D_{c,t}[i, j] = \left| \tilde{G}_t^{\text{AOI}}[i, j] - B_c[i, j] \right| \quad (7) \quad B_c[i, j] = \text{median}\left(\tilde{G}_{c,1}^{\text{AOI}}[i, j], \dots, \tilde{G}_{c,n}^{\text{AOI}}[i, j]\right) \quad (8)$$

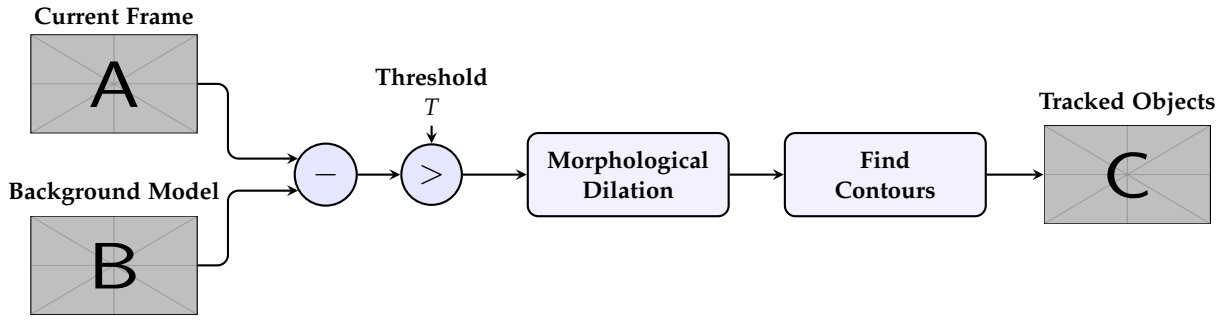


Figure 3: Background subtraction pipeline (adapted from [4]).

A binary threshold, τ was applied using a cut-off value of 50. Pixel intensities less than 50 were classified as background noise (0), while values strictly greater than or equal to 50 were designated as the foreground object (255).

$$T_t[i, j] = \begin{cases} 255 & \text{if } D_t[i, j] \geq \tau \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

While background subtraction successfully removed the static background, object detection was hindered by foreground fragmentation. Subtle variations in pixel intensity left several disconnected foreground regions. To consolidate these regions, a further morphological operation was applied. Utilising the Minkowski Sum or dilation the result, $T_t^{(2)}$ was computed:

$$T_t^{(2)} = (T_t \oplus E) \oplus E \quad (\text{where superscript 2 denotes two passes}) \quad (10)$$

$$(T \oplus E)[i, j] = \max_{(u, v) \in E} T[i + u, j + v] \quad (11)$$

Where E is a 5×5 structuring element and $\oplus$ represents the dilation operator. An elliptical structuring element was chosen as it better approximates the circular shape of the object.

Contours of the remaining contiguous regions were extracted using the Suzuki-Abe algorithm[5] via OpenCV[6]. The algorithm identified a set of boundary pixels, $[i, j]$, where foreground regions transition to the background. By joining contiguous boundary pixels of equal intensity, a set of the outermost contours was derived.

The centre, or 'centroid', of the primary foreground mask was computed using image moments. Three specific moments were utilised to compute the mask's total area (number of foreground pixels) and the sum of both the i and j coordinates within that region.

$$m_{pq} = \sum_r \sum_s r^p \cdot s^q \cdot T^{(2)}[r, s], \quad p, q \in \{0, 1\} \quad (12)$$

To calculate the centre of mass (c_x, c_y) the first moment, m_{10} was divided by the zeroth, m_{00} , essentially computing the weighted average position of each white pixel. This method successfully extracted the object's spatial centre in each frame, maintaining a computational efficiency of $\mathcal{O}(n)$.

$$c_x = \frac{m_{10}}{m_{00}}, \quad c_y = \frac{m_{01}}{m_{00}} \quad (13)$$

2.3 Time-Frequency Analysis

Following centroid extraction, time-series plots were generated to visualise oscillations along the x and y axes for each camera. Due to variability in the maximum number of frames between cameras, each dataset was truncated to the minimum number of frames observed (392). Subsequently, the

dominant oscillation frequencies for one camera (camera 3) were calculated using the tracked centroid trajectories. Camera 3 was chosen for frequency analysis as it exhibited the clearest oscillatory structure among the three cameras. To prepare for frequency analysis, the raw centroid positions, $c_x[t]$, were zero-centred by subtracting the mean, thereby removing the DC offset:

$$\tilde{c}_x[t] = c_x[t] - \frac{1}{n} \sum_{k=0}^{n-1} c_x[k] \quad (14)$$

where n is the total number of frames. The Fast Fourier Transform (FFT) was then applied to the mean-subtracted signal:

$$X[k] = \sum_{t=0}^{n-1} \tilde{c}_x[t] e^{-i2\pi kt/n}, \quad k = \frac{1}{n} \cdot \left[-\frac{n}{2}, \dots, -1, 0, 1, \dots, \frac{n}{2} - 1 \right] \quad (15)$$

where k is the frequency variable, in cycles/frame (centred around zero using `fftshift`). Positive frequency components were isolated using a mask ($f_k > 0$), and dominant frequencies identified as the maximum magnitude over the positive frequency range:

$$k^* = \arg \max_{k: f_k > 0} |X[k]| \quad (16)$$

To analyse the evolution of the frequency components over time, a Short-Time Fourier Transform (STFT) was applied to the trajectories using a Gaussian window function, known as the Gábor Transform:

$$\mathcal{G}\{\tilde{c}_x\}(\tau, f) = \sum_{t=0}^{N-1} \tilde{c}_x[t] g(t - \tau) e^{-i2\pi ft} \quad (17)$$

where τ is the centre of the sliding window and $g(t - \tau)$ is the Gaussian kernel, defined as:

$$g(t - \tau) = \exp\left(-\frac{(t - \tau)^2}{a^2}\right) \quad (18)$$

where the spread of the window is σ , initialised at 60. Several window widths ($a = [20, 40, 60, 100]$) were tested in accordance with the frequency-time trade-off presented in Gabor's Uncertainty Principle, which built upon the work by Heisenberg in the realm of quantum mechanics[7]. The theorem states that lower window widths provide good time resolution but poor frequency resolution, and vice versa:

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi} \quad (19)$$

At each window centre τ_i , the windowed signal was Fourier transformed and the normalised magnitude spectrum stored:

$$S(\tau_i, f) = \frac{|\mathcal{G}\{\tilde{c}_x\}(\tau_i, f)|}{\max_f |\mathcal{G}\{\tilde{c}_x\}(\tau_i, f)|} \quad (20)$$

The resulting spectrogram was plotted to identify non-linearities in the system behaviour during the oscillation frequency.

2.4 Dimensionality Reduction & Analytical Modelling

Following the motion tracking, the oscillating object's dynamics were captured across three camera angles, yielding six variables for each target position. Due to the large amounts of overlap in the tracked data, many of the variables were highly correlated and, therefore, could be made redundant. To recover the true underlying degrees of freedom of the hybrid spring-pendulum system,

Principal Component Analysis (PCA) was utilised to extract the dominant modes within the high dimensional dataset. The system is expected to lie within a low-dimensional manifold embedded within $\mathbb{R}^6$, corresponding to the pendulum swing and vertical spring oscillation. Typically, PCA involves computing the eigenvectors and eigenvalues of the covariance matrix, whereas SVD deals only with singular values. Rather than direct eigendecomposition, sklearn's implementation employs SVD (Singular Value Decomposition) which reduces numerical instability related to floating point precision.

Prior to constructing the data matrix for PCA, each centroid vector was independently passed through a Power Spectral Density (PSD) filter. This step was introduced to suppress noise created by the object detection pipeline, such as inconsistencies in determining an accurate centroid. If left un-filtered, the input data could significantly reduce the precision of the explained variance from the PCA. Each centroid vector was passed through a Discrete Fourier Transform before computing the PSD for each frequency bin:

$$\text{PSD}[k] = |S[k]|^2 \quad (21)$$

A binary mask was then applied, retaining only components whose power exceeded a threshold, α of 0.1. The signal was then transformed back to the time-domain using an Inverse Fourier Transform:

$$\tilde{S}[k] = \begin{cases} S[k] & \text{if } |S[k]|^2 > \alpha \cdot \max_k |S[k]|^2 \\ 0 & \text{otherwise} \end{cases} \quad (22) \quad \tilde{s}[t] = \mathcal{F}^{-1}\{\tilde{S}[k]\} \quad (23)$$

Following PSD filtering, the six mean-centred centroids were combined into a data matrix $\mathbf{X} \in \mathbb{R}^{n \times 6}$, where n corresponds to the minimum number of frames across each camera (392):

$$\mathbf{X} = [\hat{c}_{x1} \ \hat{c}_{y1} \ \hat{c}_{x2} \ \hat{c}_{y2} \ \hat{c}_{x3} \ \hat{c}_{y3}] \quad (24)$$

Prior to PCA, the data matrix was unit-variance scaled. This was essential to reduce the impact of large raw variances across different cameras, which the PCA would assign disproportionately high weights to. This standardisation technique ensures each coordinate has an equal contribution to the covariance structure.

As mentioned previously, the implementation of PCA within `sklearn.decomposition.PCA`, does not perform eigendecomposition directly on the covariance matrix $\mathbf{C}$, but it in fact computes the economy SVD of $\mathbf{X}$:

$$\mathbf{X}' = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \quad (25)$$

where $\mathbf{U} \in \mathbb{R}^{n \times p}$ and $\mathbf{V} \in \mathbb{R}^{p \times p}$ are the left and right singular vectors of $\mathbf{X}$, and $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_p)$ represents the matrix of singular values. The explained variances of each principal component were plotted on a scree plot, using the elbow method and a threshold of 70% to determine the number of principal components to retain. The dominant frequencies of the extracted principal components were obtained via FFT. This was converted into angular frequency, ω_i (rad/s) using an estimated frame rate of 24 fps. As the dataset metadata did not include a sampling rate, this was inferred based on the observed oscillations. The length of the pendulum was computed using the angular frequency derived from PC1 and the spring constant, k from PC2 using the following equations:

$$k = m\omega_1^2, \quad L = \frac{g}{\omega_2^2} \quad (26)$$

where m is the estimated mass of the object (0.5 kg), estimated based the typical weight of an empty paint can and g is the acceleration due to gravity (9.81 m/s²). These estimated variables, derived from Lagrangian mechanics[3] were used to validate the experimental data by overlaying the analytical sinusoidal waves $A_i \cos(\omega_i t)$ against the PSD filtered PC1 and PC2 data.

$$\theta(t) = A_\theta \cos(\omega_\theta t + \phi_0), \quad z(t) = A_z \cos(\omega_z t + \phi_z) \quad (27)$$

3 Results

The tracking pipeline, outlined in **Figure 2** and **3** were applied to each camera frame successfully. For each frame, the foreground mask correctly removed the static background, and the morphological techniques accurately determined the contours of the mass and its corresponding coordinates. The final mean-subtracted centroid tracking results are shown in **Figure 4**.

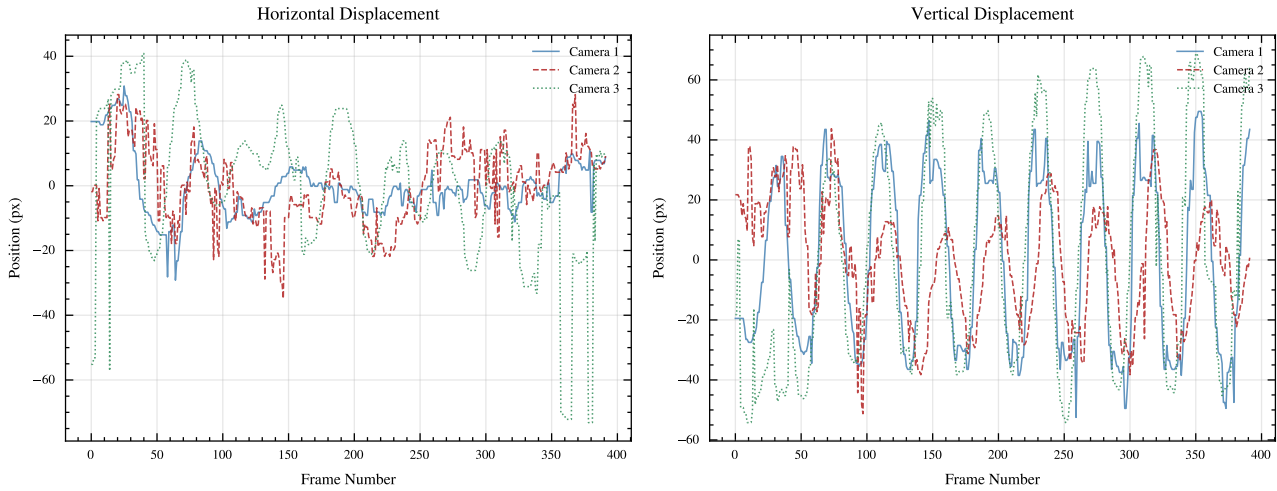


Figure 4: Mean-subtracted centroid trajectories for each camera: Horizontal displacement, c_x (left) & Vertical displacement, c_y (right).

Each camera exhibited clear oscillatory motion in both the x and y axes. Additionally, each signal exhibits a gradual decay, indicating both energy exchange between modes and overall system damping. Dominant frequencies were extracted for Camera 3's c_x and c_y coordinates using FFT. After isolating the positive frequency components, the horizontal and vertical signals both yielded dominant frequencies of approximately 0.0255 cycles/frame corresponding to the pendulum and spring oscillation. The identical dominant frequencies in both axes likely reflects Camera 3's viewing axis, which overlaps both modes onto each others pixel axes, obscuring the pendulum swing.

The Gábor spectrogram (**Figure 5**) reveals a stable and persistent frequency band at approximately 0.0255 cycles/frame for both vertical and horizontal axes of Camera 3. Furthermore, minimal drift was observed in the frequency components, similar to that of a weakly-damped oscillator.

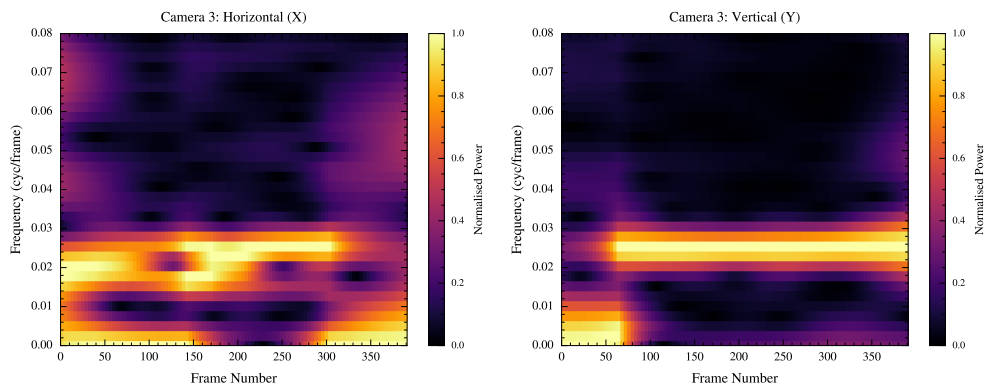


Figure 5: Gabor Spectrogram for Camera 3 (Window Width = 60).

PCA was applied to the combined data matrix of PSD-filtered signals from all three cameras, following standardisation scaling. The resulting scree plot and cumulative explained variance are shown in **Figure 6**.

The first three principal components account for 80.9% of the total variance, with the first principal component comprising 40.7%. The remaining components account for 19.1%, concluding that

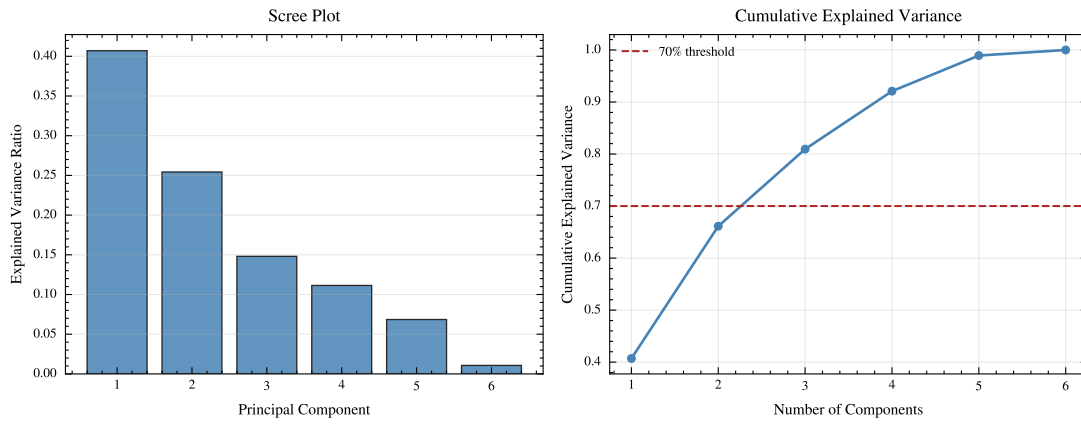


Figure 6: Scree plot of individual explained variance ratios (left) and cumulative explained variance with 70% threshold (right)

the system dynamics are largely driven by 2–3 modes. The PC loadings expressed large contribution of the vertical coordinates (c_{y1}, c_{y2}, c_{y3}) to PC1 and horizontal signals (c_{x1}, c_{x2}, c_{x3}) to PC2, demonstrating that the two dominant modes correlate to the vertical, spring oscillation (PC1) and horizontal pendulum motion (PC2). PC3 captures 14.8% of the residual variance, likely accounting for additional vertical oscillation or system motion produced by releasing the mass with an initial velocity perpendicular to the equilibrium position. The dominant frequencies of the first two principal components were 0.0255 and 0.0026 cycles/frames, respectively. PC2's initial dominant frequency however corresponded to the minimal observable frequency bin and in fact the correct value lay closer to 0.025 cycles/frame. The similarities between the two natural frequencies is consistent with the observed frequencies and energy exchange found in centroid tracking and frequency analysis.

Taking the observed dominant frequencies for PC1 and PC2 an analytical model was created, estimating the pendulum's length, L as 0.69 m and spring constant, k as 7.4 N/m. The resulting sinusoidal waves were plotted against the experimental modes, showing a close frequency component between the spring mode (PC1) and the analytical model, with a slight drift beginning at 8 seconds due to non-linear coupling effects. Conversely, the analytical model did not replicate the energy exchange exhibited in PC2's amplitude variation but mirrored that produced by the spring component. This is confirmed by their associated RMSE values of 1.15 and 2.34 respectively.

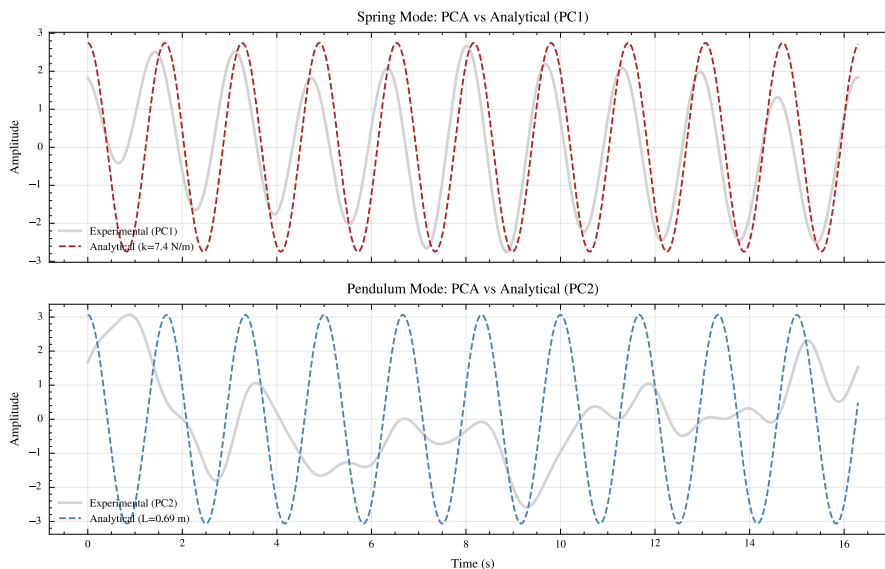


Figure 7: Comparison of temporal PCA signals and analytically derived model.

4 Discussion

The results indicate that there were several pitfalls emerging from the pre-processing and motion tracking pipeline. Initial image processing steps including median background subtraction were effective at isolating the oscillating mass. However, the motion tracking and contour detection techniques did not accurately measure the mass centroid across each frame. These inaccuracies were largely due to variations in colour, as the mass rotated and remaining contours of surrounding objects which were often mistaken for the target mass. The result was a system highly sensitive to large vertical oscillations, yet inadequate for detecting genuine horizontal pendulum motion. Additionally, AOI identification was completed manually and therefore, not generalisable to other hybrid systems.

The Gábor spectrogram displayed a stable band near the vertical system's natural frequency for both centroid vectors (c_x and c_y), indicating energy level stability and mode convergence. Critically, both the modes share the same dominant frequency of 0.0255 cycles/frame. This highlights a primary drawback of time-frequency analysis alone: the resulting spectrograms cannot sufficiently describe mode coupling nor do they explain the mechanisms of energy transfer between these modes. Consequently, the choice to apply PCA offers an orthogonal approach, which instead analyses the frequency components across each camera angle.

The resulting explained variance aligns with the expected modes of the system. Accounting for 40.7% and 25.4% respectively, the concentration of variance within these two components is consistent with a system comprising of two dominant modes of motion. By analysing the loading coefficients, we see a large contribution to PC1 from vertical motion indicating a spring motion, whilst PC2 loadings comprised of largely horizontal data, identifying it as the pendulum mode. One discrepancy noted was the disproportionately large correlation of Camera 3's c_x to PC1. It is my belief that this is due to the optical axis of this camera, which if aligned to the pendulum's swing mode, could make it sensitive to the spring extension dynamics rather than the swing.

A further limitation merged in the dominant frequency extraction of PC2. Initial Fourier analysis determined a large peak at 0.0026 cycles/frame which corresponded to the minimum frequency bin ($\Delta f = 1/392$), representing a DC-drift rather than the true oscillation frequency. After excluding the DC components the true frequency was found to be 0.025 cycles/frame which closely matched that of PC1. The corrected analytical parameters, $k = 7.4 \text{ N/m}$ and $L = 0.69 \text{ m}$, were then far more reasonable. Furthermore, the lack of camera calibration may have distorted PCA results to certain modes of motion. The analytical comparison (shown in **Figure 7**) assessed the validity of the decoupled model. The spring mode (PC1) exhibited strong correlation to the analytical, un-damped cosine wave with a slight phase drift emerging towards the middle half of the frames. Conversely, the pendulum mode (PC2) had significantly poorer interaction with the linear model. This confirms the decoupled model's inability to accurately model the complex non-linear dynamics and mode coupling present in a hybrid dynamical system.

5 Conclusion

This report analysed a hybrid elastic pendulum system using image processing, computer vision and dimensionality reduction methods. Background subtraction and centroid extraction determined the object's trajectory from each camera, and Fourier analysis confirmed a stable, dominant frequency at 0.0255 cycles/frame. PCA reduced the dimensionality of the data and identified two dominant modes; vertical, spring oscillation (PC1, 40.7%) and horizontal, pendulum swing (PC2, 25.4%). Physical system parameters were estimated as $k = 7.4 \text{ N/m}$ and $L = 0.66 \text{ m}$ which formed the basis for the decoupled analytical model. The linearised model closely matched the oscillations of the spring mode but failed to capture the coupling dynamics exhibited in the pendulum mode. Further improvement could be made to the experimental setup by using a camera mount to reduce motion blur and a high-contrast colour marker on the oscillating mass.

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Appendix A: Analytical Model Derivation

Adapted from [3].

Lagrangian Formulation

$$x = r \sin \theta, \quad z = r \cos \theta \quad (28)$$

$$T = \frac{1}{2}m (\dot{r}^2 + r^2\dot{\theta}^2) \quad (29)$$

$$V = \frac{1}{2}k(r - L_0)^2 - mgr \cos \theta \quad (30)$$

$$\mathcal{L} = \frac{1}{2}m (\dot{r}^2 + r^2\dot{\theta}^2) - \frac{1}{2}k(r - L_0)^2 + mgr \cos \theta \quad (31)$$

Full Nonlinear Equations of Motion

Radial (r) equation:

$$m\ddot{r} = mr\dot{\theta}^2 - k(r - L_0) + mg \cos \theta \quad (32)$$

Angular (θ) equation:

$$r\ddot{\theta} + 2\dot{r}\dot{\theta} + g \sin \theta = 0 \quad (33)$$

Equilibrium and Linearisation

$$k(r_{\text{eq}} - L_0) = mg \quad \implies \quad r_{\text{eq}} = L_0 + \frac{mg}{k} \quad (34)$$

$$\sin \theta \approx \theta, \quad \cos \theta \approx 1, \quad \dot{\theta}^2 \approx 0, \quad \dot{r}\dot{\theta} \approx 0 \quad (35)$$

Linearised radial equation:

$$m\ddot{\Delta r} = -k \underbrace{(r_{\text{eq}} - L_0)}_{= mg/k} - k\Delta r + mg = -mg - k\Delta r + mg \quad (36)$$

$$\boxed{\ddot{\Delta r} + \frac{k}{m}\Delta r = 0} \quad \omega_{\text{spring}} = \sqrt{\frac{k}{m}} \quad (37)$$

Linearised angular equation:

$$r_{\text{eq}}\ddot{\theta} + g\theta = 0 \quad (38)$$

$$\boxed{\ddot{\theta} + \frac{g}{r_{\text{eq}}}\theta = 0} \quad \omega_{\text{pend}} = \sqrt{\frac{g}{r_{\text{eq}}}} \quad (39)$$